

FFAR Final Progress Report

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As part of closing out a FFAR grant, grantees must complete a Final Progress Report. Please use the template below to complete the programmatic report. The report must be submitted to FFAR within 90 days after the expiration or termination of a FFAR funded research grant. All questions about this form should be directed to grants@foundationfar.org.

The Final Progress Report communicates the cumulative results and major accomplishments of the funded grant research, including accomplishment for each aim and whether all research aims were fully completed. FFAR uses the content of the Final Progress Report to inform Congress, the Advisory Board, and other stakeholders on the successes, impacts and value of funding unique partnerships to support innovative science addressing today's food and agriculture challenges. If a question is not applicable to the project, please type "Nothing to report."

Grant Information	
Grant ID	DFs-18-0000000012
Award Program	Direct Solicitation Award
Project Title	Ecosystem Services Market Consortium
Grant Start Date	01/01/2019
Grant End Date	1/31/2023
Total Grant Budget	\$20,600,000
Total FFAR Budget	\$10,300,000
Matching Funder(s)	\$10,300,000
Final Report Date	1/31/2023
Project Director/Principal Investigator Information	
Full Name	Debbie Reed
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Grantee Organization Information	
Organization Name	Ecosystem Services Market Consortium
Address	3426 Greentree Drive
City, State Zip	Falls Church, VA 22041
Tax ID	86-1877472
Authorized Signing Official (ASO) Information	
Name	Debbie Reed
ASO Title	Executive Director
Email	dreed@ecosystems-servicesmarket.org
Phone	202-701-4298

1. GENERAL INFORMATION

1.1. Please list the geographic location(s) – city, state, congressional district - where the work was conducted. If the work was conducted outside of the U.S., please list the city and country.

Illinois

- 1 Rep. Jonathan Jackson (D)
- 2 Rep. Robin Kelly (D)
- 3 Rep. Delia Ramirez (D)
- 4 Rep. Jesús García (D)
- 5 Rep. Mike Quigley (D)
- 6 Rep. Sean Casten (D)
- 7 Rep. Danny Davis (D)
- 8 Rep. Raja Krishnamoorthi (D)
- 9 Rep. Janice Schakowsky (D)
- 10 Rep. Bradley Schneider (D)
- 11 Rep. Bill Foster (D)
- 12 Rep. Mike Bost (R)
- 13 Rep. Nikki Budzinski (D)
- 14 Rep. Lauren Underwood (D)
- 15 Rep. Mary Miller (R)
- 16 Rep. Darin LaHood (R)
- 17 Rep. Eric Sorensen (D)

Indiana

- 1 Rep. Frank Mrvan (D)
- 2 Rep. Rudy Yakym (R)
- 3 Rep. Jim Banks (R)
- 4 Rep. James Baird (R)
- 5 Rep. Victoria Spartz (R)
- 6 Rep. Greg Pence (R)
- 7 Rep. André Carson (D)
- 8 Rep. Larry Bucshon (R)
- 9 Rep. Erin Houchin (R)

Iowa

- 1 Rep. Mariannette Miller-Meeks (R)
- 2 Rep. Ashley Hinson (R)
- 3 Rep. Zach Nunn (D)
- 4 Rep. Randy Feenstra (R)

Kansas

- 1 Rep. Tracey Mann (R)
- 2 Rep. Jacob LaTurner (R)
- 3 Rep. Sharice Davids (D)
- 4 Rep. Ron Estes (R)

Michigan

- 1 Rep. Jack Bergman (R)
- 2 Rep. John Moolenaar (R)
- 3 Rep. Hillary Scholten (D)
- 4 Rep. Bill Huizenga (R)
- 5 Rep. Tim Walberg (R)
- 6 Rep. Debbie Dingell (D)
- 7 Rep. Elissa Slotkin (D)
- 8 Rep. Dan Kildee (D)
- 9 Rep. Lisa McClain (R)
- 10 Rep. John James (R)
- 11 Rep. Haley Stevens (D)
- 12 Rep. Rashida Tlaib (D)
- 13 Rep. Shri Thanedar (D)

Minnesota

- 1 Rep. Brad Finstad (R)
- 2 Rep. Angie Craig (D)
- 3 Rep. Dean Phillips (D)
- 4 Rep. Betty McCollum (D)
- 5 Rep. Ilhan Omar (D)
- 6 Rep. Tom Emmer (R)
- 7 Rep. Michelle Fischbach (R)
- 8 Rep. Pete Stauber (R)

Missouri

- 1 Rep. Cori Bush (D)
- 2 Rep. Ann Wagner (R)
- 3 Rep. Blaine Luetkemeyer (R)
- 4 Rep. Mark Alford (R)
- 5 Rep. Emanuel Cleaver II (D)
- 6 Rep. Sam Graves (R)
- 7 Rep. Eric Burlison (R)
- 8 Rep. Jason Smith (R)

Nebraska

- 1 Rep. Mike Flood (R)
- 2 Rep. Don Bacon (R)
- 3 Rep. Adrian Smith (R)

North Dakota

- At-Large Rep. Kelly Armstrong (R)

Ohio

- 1 Rep. Greg Landsman (D)
- 2 Rep. Brad Wenstrup (R)
- 3 Rep. Joyce Beatty (D)
- 4 Rep. Jim Jordan (R)
- 5 Rep. Robert Latta (R)
- 6 Rep. Bill Johnson (R)
- 7 Rep. Max Miller (R)
- 8 Rep. Warren Davidson (R)
- 9 Rep. Marcy Kaptur (D)
- 10 Rep. Michael Turner (R)
- 11 Rep. Shontel Brown (D)
- 12 Rep. Troy Balderson (R)
- 13 Rep. Emilia Sykes (D)
- 14 Rep. David Joyce (R)
- 15 Rep. Mike Carey (R)

Oklahoma

- 1 Rep. Kevin Hern (R)
- 2 Rep. Josh Brecheen (R)
- 3 Rep. Frank Lucas (R)
- 4 Rep. Tom Cole (R)
- 5 Rep. Stephanie Bice (R)

South Dakota

- At-Large Rep. Dusty Johnson (R)

Tennessee

- 1 Rep. Diana Harshbarger (R)
- 2 Rep. Tim Burchett (R)
- 3 Rep. Charles “Chuck” Fleischmann (R)
- 4 Rep. Scott DesJarlais (R)
- 5 Rep. Andy Ogles (R)
- 6 Rep. John Rose (R)
- 7 Rep. Mark Green (R)
- 8 Rep. David Kustoff (R)
- 9 Rep. Steve Cohen (D)

Wisconsin

- 1 Rep. Bryan Steil (R)

- 2 Rep. Mark Pocan (D)
- 3 Rep. Derrick Van Orden (R)
- 4 Rep. Gwen Moore (D)
- 5 Rep. Scott Fitzgerald (R)
6. Rep. Glenn Grothman (R)
- 7.Rep. Thomas P. Tiffany (R)
- 8.Rep. Mike Gallagher (R)

Texas

1. Rep. Nathaniel Moran (R)
2. Rep. Dan Crenshaw (R)
3. Rep. Keith Self (R)
4. Rep. Patrick Fallon (R)
5. Rep. Lance Gooden (R)
6. Rep. Jake Ellzey (R)
7. Rep. Lizzie Fletcher (D)
8. Rep. Morgan Luttrell (R)
9. Rep. Al Green (D)
10. Rep. Michael McCaul (R)
11. Rep. August Pfluger II (R)
12. Rep. Kay Granger (R)
13. Rep. Ronny Jackson (R)
14. Rep. Randy Weber Sr. (R)
15. Rep. Monica De La Cruz (R)
16. Rep. Veronica Escobar (D)
17. Rep. Pete Sessions (R)
18. Rep. Sheila Jackson Lee (D)
19. Rep. Jodey Arrington (R)
20. Rep. Joaquin Castro (D)
21. Rep. Chip Roy (D)
22. Rep. Troy Nehls (R)
23. Rep. Ernest 'Tony' Gonzales (R)
24. Rep. Beth Van Duyne (R)
25. Rep. Roger Williams (R)
26. Rep. Michael Burgess (R)
27. Rep. Michael Cloud (R)
28. Rep. Henry Cuellar (D)
29. Rep. Sylvia Garcia (D)
30. Rep. Jasmine Crockett (D)
31. Rep. John Carter (R)
32. Rep. Colin Allred (D)
33. Rep. Marc Veasey (D)
34. Rep. Vicente Gonzalez (D)
35. Rep. Greg Casar (D)
36. Rep. Brian Babin (R)

Oregon

1. Rep. Suzanne Bonamici (D)

2. Rep. Cliff Bentz (R)
3. Rep. Earl Blumenauer (D)
4. Rep. Val Hoyle (D)
5. Rep. Lori Chavez-DeRemer (D)
6. Rep. Andrea Salinas (D)

California

1. Rep. Doug LaMalfa (R)
2. Rep. Jared Huffman (D)
3. Rep. Kevin Kiley (R)
4. Rep. Mike Thompson (D)
5. Rep. Tom McClintock (R)
6. Rep. Ami Bera (D)
7. Rep. Doris Matsui (D)
8. Rep. John Garamendi (D)
9. Rep. Josh Harder (D)
10. Rep. Mark DeSaulnier (D)
11. Rep. Nancy Pelosi (D)
12. Rep. Barbara Lee (D)
13. Rep. John Duarte (R)
14. Rep. Eric Swalwell (D)
15. Rep. Kevin Mullin (D)
16. Rep. Anna Eshoo (D)
17. Rep. Ro Khanna (D)
18. Rep. Zoe Lofgren (D)
19. Rep. Jimmy Panetta (D)
20. Rep. Kevin McCarthy (R)
21. Rep. Jim Costa (D)
22. Rep. David Valadao (R)
23. Rep. Jay Obernolte (R)
24. Rep. Salud Carbajal (R)
25. Rep. Raul Ruiz (D)
26. Rep. Julia Brownley (D)
27. Rep. Mike Garcia (R)
28. Rep. Judy Chu (D)
29. Rep. Tony Cárdenas (D)
30. Rep. Adam Schiff (D)
31. Rep. Grace Napolitano (D)
32. Rep. Brad Sherman (D)
33. Rep. Pete Aguilar (D)
34. Rep. Jimmy Gomez (D)
35. Rep. Norma Torres (D)
36. Rep. Ted Lieu (D)
37. Rep. Sydney Kamlager-Dove (D)
38. Rep. Linda Sánchez (D)
39. Rep. Mark Takano (D)
40. Rep. Young Kim (R)
41. Rep. Ken Calvert (R)

- 42. Rep. Robert Garcia (D)
- 43. Rep. Maxine Waters (D)
- 44. Rep. Nanette Barragán (D)
- 45. Rep. Michelle Steel (R)
- 46. Rep. Lou Correa (D)
- 47. Rep. Katie Porter (D)
- 48. Rep. Darrell Issa (R)
- 49. Rep. Mike Levin (D)
- 50. Rep. Scott Peters (D)
- 51. Rep. Sara Jacobs (D)
- 52. Rep. Juan Vargas (D)

1.2 How many new jobs were created by the grant during this reporting period?

None.

1.3 How many jobs were maintained by the grant during this reporting period?

Six key and additional other personnel roles were maintained. We also have contracted with 42 vendors, and the majority of those contracts support multiple resources.

1.4 Have there been any changes to your organization's IRS 501(c)(3) non-profit status since you were awarded the grant?

This grant was originally awarded to the Soil Health Institute (SHI) which acted as fiscal sponsor for ESMC LLC. As of September 1, 2021, with agreement of FFAR, SHI and ESMC/ESMRC, the award was transferred to Ecosystem Services Market Consortium, a Virginia non-stock corporation granted 501(c)(3) exempt status by the IRS on June 4, 2021.

1.5 Has your organization undergone a recent name change? If so, please provide the new name of your organization.

Please refer to the previous answer.

2. Scientific Report

2.1 Public Abstract (up to 500 words; current word count = 499): The abstract should be 'stand-alone' and is intended for a general audience. It should be a concise overview/summary of the importance of the project, the issues that the project addressed, and the key findings of the project. The abstract should also clearly state how the results of the project help to address important needs in U.S. food and agriculture systems.

The Ecosystem Services Market Consortium (ESMC) was established to undertake ground-breaking research, development, demonstration and deployment activities to launch an innovative ecosystem services market program dedicated to scaling beneficial climate and natural resource outcomes from US agriculture. Through an innovative consortium model

comprised of more than 70 member organizations spanning the agricultural supply and value chains, collaborative investments have been made in developing, testing, and launching a standardized market aimed at decarbonizing the agricultural supply chain.

In May 2022, ESMC launched its market program, Eco-Harvest, to quantify and verify carbon, GHG and water outcomes annually and to pay farmers from the sale of credits to corporates. At the heart of Eco-Harvest is ESMC's Monitoring, Reporting, and Verification (MRV) platform, a technologically advanced, end-to-end digitized solution that generates high quality, third-party accredited and verified Scope 3 supply chain credits for soil carbon removals and reduced greenhouse gas emissions. ESMC/ESMRC's business model and program is the first globally to generate these outcomes in a way that meets corporate reporting requirements and the first to be validated and verified by SustainCERT.

ESMC's collaborative market design ensures that we meet the needs of farmers and ranchers ("sellers") and corporates ("buyers"), as well as critical stakeholders by improving climate, environment, and natural resource outcomes from US agriculture. We are the first program globally to have integrated and delivered water quality and water quantity impacts into our market program and have piloted approaches to incorporating biodiversity outcomes. Over the course of this project, nearly 200 participating producers in the US have benefited from improved operational resilience resulting from the implementation of eligible practice changes. Corporate investors in the food and beverage sector have benefited through a scaled market program that can deliver quantified, verified, high integrity emissions reductions and increased carbon sequestration and natural resource impact solutions for their agricultural supply chains.

ESMC's research arm, the Ecosystem Services Market Research Consortium (ESMRC), has invested in innovative, high-priority research, development, demonstration, and deployment (RDD&D) activities to overcome market challenges and continuously improve program functions, operations, and impact at scale. Central to ESMRC's research priorities has been the development and enhancement of scientifically rigorous, cost-effective soil carbon and water quality/quantity quantification approaches used to inform crediting outcomes for participating producers. Other research priorities over the course of this project included evaluating the functional capabilities and commercial viability of remote-sensing and proximal-sensing soil sampling technologies to more efficiently and cost effectively measure and quantify soil carbon and net GHG changes, as well as remotely-sensed data products derived from satellite imagery to improve both the cost and accuracy of collecting field-level data for outcomes quantification; refining the APEX model to accommodate additional conservation practices for tile drainage; and exploring approaches to soil sampling that appropriately balance measurement accuracy and cost.

Collectively, these efforts provide those operating in the ecosystem services market space with new insights and mechanisms to accelerate the viability of programs that reward producers for

implementing management practices that result in positive environmental outcomes for all of society.

- 2.2. Provide a description or interpretation of how the key findings of the project can be used by the agriculture community. It is important the PI identifies how these findings have been, or may be, adopted or adapted in U.S. agriculture and food systems. If the findings cannot yet be applied, this section should address how they can be used to guide future planning or decision-making (up to 1,000-word limit; current word count = 738).

The core result of this project is the launch of the first fully-accredited scope 3 ecosystem services market program in the country, and the only one with an integrated carbon, water and biodiversity capability. ESMC's Eco-Harvest market program launched in May 2022. The supporting Monitoring, Reporting, and Verification (MRV) platform is an end-to-end digitized and advanced technology program and platform that generates high quality, third-party verified scope 3 credits for soil carbon removals, reduced greenhouse gases, and water impacts from U.S. agriculture. Eco-Harvest establishes clear pathways of engagement in ecosystem service market and connects producers to ESMC's on-the-ground partners for technical and agronomic support. The Eco-Harvest platform overcomes past market challenges with a technologically advanced infrastructure that is scalable, replicable, and cost-effective. Program documentation created to outline market program guidelines, eligibility requirements, the enrollment process, and eligible program regions is intended for use by project partners and interested producers to facilitate their participation in Eco-Harvest.

To enhance the experience of project partners and producers participating in our program, ESMC specifically designed the MRV to streamline producer data collection and to automate QA/QC procedures and gap-filling to generate data for modeling, with the goal of reducing data entry burden and errors. Substantial investments were made to automate both core and ancillary functions of the MRV, including gap-filling; data QA/QC; water quality and water quantity outcomes quantification; pre- and post-model processing; and tracking, reporting, and preparing producer data to transfer to the S3S tool for credit generation. This first-of-its kind data collection and processing system will directly benefit the food and agricultural community by reducing the cost of quantification and outcomes generation and ensuring the generation of high-quality, verified Scope 3 supply chain credits for ecosystem services that deliver value to producers, corporate buyers, and society at large in the effort to meet net zero carbon and greenhouse gas commitments and corporate ESG goals.

This project also supported numerous investments in the public good to accelerate improvements to U.S. agriculture and food systems. Through the FFAR grant, ESMC funded the addition of four conservation practice modules for tile drainage to the APEX model. Source code was modified to provide modeling capacity for drainage water management, bioreactors, constructed wetlands, and saturated buffers. These enhancements are publicly available to anyone who uses APEX to identify best management practices, find sources of pollutants that affect water quality, and manage water resources on croplands. Substantial investments were also made in the open-source Stratification App, a web-based application that generates highly accurate soil carbon sampling and monitoring plans and provides real-time, in-field navigation

and soil collection support. Additionally, this project funded the Advanced Quantification Technology Assessment Project, a substantial body of research that provides insight into the functional capabilities, commercial viability, and market-readiness of both remote-sensing and proximal-sensing soil sampling technologies to more efficiently and cost effectively measure and quantify soil C and net GHG changes, as well as remotely-sensed data products derived from satellite imagery to improve both the cost and accuracy of collecting field-level data for outcomes quantification (the Comparative Analysis of Remote Sensing Data Solutions Project). Reports from the Advanced Quantification Technology Assessment Project are publicly available on ESMC's website to support the community's decision-making around the use of alternative technologies for soil sampling and field-data collection. The Comparative Analysis of Remote Sensing Data Solutions report will be released to the public in January 2024, per ESMC's policy to retain reports for members only for twelve months upon completion. Soil Carbon Technology Testing and Training Data Sets were also developed to support calibration and improved accuracy of remote sensing technologies that predict soil organic carbon (SOC) and/or bulk density (BD). These data sets will be made available to tool providers via an automated testing service, to enable predictions of SOC and BD and inform refinements to prediction algorithms.

Lastly, insights from the Assessment of Ecosystem Service Marketplace Programming project can be leveraged by the broader ecosystem services market community to better understand the factors that project partners and producers consider when determining whether to participate in a payment-for-ecosystem-service program. To foster a positive experience, such programs must be designed to minimize and/or streamline data collection, adequately compensate for the role of project coordinators and enrollment specialists and deliver maximum compensation to producers for the value they bring to corporations and society through ecosystem services. ESMC is constantly adapting and innovating to provide the most robust and rewarding end-to-end experience for those participating in the Eco-Harvest program. Further evaluation will be undertaken as the program evolves to determine how this initiative shapes the future of U.S. agriculture and food systems.

- 2.3 Research Outcomes/Impact (up to 1,500-word limit; current word count = 863). The primary goal here is to answer questions such as “How did this project lead to improvements in U.S. agriculture and food systems?” or “How can the findings of this study guide or make improvement in future investigations and research?” Outcomes should be explained and classified in one of the following ways:
- a) potential outcomes, i.e., findings, results, or recommendations that could impact U.S. agriculture and food systems, if used;
 - b) b. intermediate outcomes, i.e., how the findings, results, or recommendations have been used by others to influence practices, legislation, research/product design, training and so forth; and
 - c) c. end outcomes, i.e., how findings, results, or recommendations have contributed to documented reductions in work-related morbidity, mortality, and/or exposure.

The legacy of the FFAR grant and the ESMC/ESMRC consortium is an advanced, functioning, and scalable full-service ecosystem services market program (Eco-Harvest) and national scale infrastructure in continued development that will lead to improved, quantified and verified ecosystem services outcomes from US agriculture at scale. Eco-Harvest, as envisioned in the original FFAR grant proposal, goals and objectives, is an end-to-end digitized and advanced technology program and platform that generates high quality, third-party accredited and verified Scope 3 supply chain credits. The program successfully developed this approach to overcome past and current market program challenges and is being widely emulated by other carbon offset, inset, and Scope 3 supply chain programs seeking to learn from our work and to establish similar programs. ESMC considers this a hallmark of success, in that more programs will help to achieve greater scale of impacts in shorter timeframes – all of which is necessary to combat the visible and escalating impacts of climate change and the negative impacts to US agricultural productivity and resilience. Due to the level of rigor in Eco-Harvest, supply chain emissions factors (EF)/impact units (IU) generated and verified through our program can be shared by others in agricultural supply chains. This allows multiple corporate partners to invest in the outcomes in a credible, verified program that ensures accuracy of corporate claims.

At the core of the Eco-Harvest Program is ESMC's Monitoring, Reporting, and Verification (MRV) Platform, which was designed to collect, validate, and process data for typical crop production scenarios. The MRV streamlines producer data collection and automates QA/QC procedures and gap-filling to generate data for modeling, with the goal of reducing data entry burden and errors. Foundational processes include: 1) collect accurate data for the parameters determined by Regrow's DNDC input and any program or certification requirements; 2) validate the accuracy and eligibility of field boundaries, specific data points entered, evidence of a practice change, and overall logic and timing of agronomic events based on a defined reporting year for a specific crop being produced; 3) implement creation of comparison scenarios of practice change vs baseline based on known standards, quantification protocols, and agricultural science that are specific to the crop being produced; and 4) send comparison scenarios in a format compatible with the DNDC input constraints/requirements so that valid outcomes can be generated. The MRV productizes the ESMC Protocol and includes core functionality such as a landing page with programmatic information, MRV enrollment, and supporting documentation for program participation; administrative functions for managing and tracking projects; field identification and validation that includes multiple visual data layers created from external sources such as CDL and PADUS; integration of AdobeSign for completing Producer Agreements; data collection portal for gathering the data needed for modeling; database for tracking producers, fields, projects, management data, modeling outcomes, and credits generated; pre-modeling data processing modules for data validation, baseline determination, and preparation of data to transfer to modeling tools; comprehensive QA/QC steps to ensure data quality and enable third-party validation of data and processing steps; and post-modeling data processing modules for outcome calculations (including the application of uncertainty adjustments at the field- and project-levels), tracking, reporting, and preparing to transfer to the S3S tool. The MRV represents a first-of-its-kind, defensible system of data collection and processing with capabilities to evolve with a dynamic market program.

The FFAR grant also enabled both the development and enhancement of scientifically rigorous and cost-effective soil carbon and water quality/quantity quantification approaches. These quantification improvements have a direct impact on our evolving understanding of the

relationship between field-level management practices and ecosystem services outcomes, as well as the value of payments that producers receive for implementing those practices. Quantification enhancements improve the accuracy of outcomes generation and often translate into cost savings, which enables a greater proportion of the value of a sold ecosystem service credit to be paid out to the producer.

Further, this project supported numerous investments in the public good to accelerate improvements to U.S. agriculture and food systems. Through the FFAR grant, ESMC funded the addition of four conservation practice modules for tile drainage to the APEX model. Source code was modified to provide modeling capacity for drainage water management, bioreactors, constructed wetlands, and saturated buffers. These enhancements are publicly available to anyone who uses APEX to identify best management practices, find sources of pollutants that affect water quality, and manage water resources on croplands. Substantial investments were also made in the open-source Stratification App, a web-based application that generates highly accurate soil carbon sampling and monitoring plans and provides real-time, in-field navigation and soil collection support. Additionally, this grant funded a substantial body of research that provided insight into the functional capabilities, commercial viability, and market-readiness of both remote-sensing and proximal-sensing soil sampling technologies to more efficiently and cost effectively measure and quantify soil C and net GHG changes, as well as remotely-sensed data products derived from satellite imagery to improve both the cost and accuracy of collecting field-level data for outcomes quantification. Collectively, these research efforts provide those operating in the ecosystem services market space with new insights and mechanisms to accelerate the end-to-end viability of programs that reward producers for implementing management practices that result in positive environmental outcomes.

- 2.4. What were the goals/specific aims of the project? If the approved application lists milestones/target dates for important activities or phases for this reporting period, identify these milestones and dates, as well as show actual completion dates or the percentage of completion of milestone targets. (up to 1,500-word limit; current word count = 1,386)

The objectives of this project were to:

1. Ability to rigorously and cost-effectively quantify science- and outcomes-based impacts of ecosystem services from agriculture at multiple scales
2. Develop rigorous, secure, automated science-based methodologies to monitor, report and verify ecosystem service outcomes from agriculture at scale
3. Develop a rigorous, secure, automated, technologically advanced science-based platform for a national-scale geospatial framework to track ecosystem service market attributes from agriculture
4. Develop a national-scale gridded Land Ledger that tracks and reports ecosystem service attributes from agriculture using static and trend data at multiple geospatial scales

5. Quantify technical capacity for carbon and water quality outcomes based on soil carbon sequestration potentials by soil types, land management approaches and climate

The following goals in support of the above objectives were identified over the course of the project:

Research

- Evaluate accuracy, usability, scalability, and cost-effectiveness of advanced in-field soil sampling technologies and identify criteria for incorporation into ESMC program (Status: Initial phases complete. Continuously in progress as part of core programming.)
- Develop training and testing data sets to improve soil organic carbon and bulk density prediction accuracy of alternative soil sampling tools (Status: Initial phase complete)
- Establish timeline and protocol for soil resampling and carbon true-ups (Status: On pause, completion pending guidance from Greenhouse Gas Protocol)
- Conduct literature review of longitudinal studies of management practices that impact long-term changes in soil organic matter at various depths (Status: Incomplete due to working group decision to abandon course)
- Identify surrogate measurements for estimating field-level N₂O emissions (Status: Incomplete due to university partner failing to complete contract agreement)
- Modify APEX model to account for regionally-relevant conservation practices in pilot regions (Status: Complete)
- Identify/develop simplified water quality and water quantity quantification approaches for Scope 3 credit quantification (Status: Complete)
- Support establishment of collaborative effort to initiate and catalyze support for a national level plan for compiling and sharing agricultural data sets for calibration and validation of agricultural water and climate quantification models and tools (Status: Incomplete due to internal decision to focus limited staff resources elsewhere)
- Obtain Ag Data Transparent Certification to foster producer confidence in and use of ESMC's MRV platform (Status: In progress)

Model and Tool Development

- Develop a mobile and web-based application to generate strata and sampling points for soil sampling (Status: Complete)
- Complete DNDC model and production system validation for select systems (Status: Complete)
- Develop functionality to transmit agronomic data between ESMC MRV and DNDC (Status: Complete)
- Develop recommendations for improved field-scale prediction of N₂O emissions and quantification approach to characterize DNDC model uncertainty (Status: Incomplete due to university partner failing to complete contract agreement)
- Establish performance benchmark metrics for DNDC to evaluate accuracy of uncertainty model (Status: Complete)
- Explore feasibility and impact of using OpTIS or other remote sensing technologies for data gap filling and practice verification in cropland projects (Status: Complete)

- Explore development of DNDC outcomes scenario tool (Status: Incomplete due to technical partner staffing constraints)
- Complete initial APEX automation work (Status: Phase I complete)
- Incorporate simplified Scope 3 water quality and quantity tools into MRV (Status: In progress)

Protocol and Verification/Certification

- Develop ESMC protocol for quantifying science- and outcomes-based impacts of ecosystem services from agriculture at multiple scales for carbon, water quality, water quantity (Status: Complete)
- Develop soil sampling protocol and identify sampling density that appropriately balances measurement accuracy, change detection, and cost (Status: Complete, undergoing subsequent evaluation)
- Refine Scope 3 asset generation pathways based on guidance from standard-setting bodies
- Complete project design validation and asset verification for VCI cropland projects (Status: Complete)
- Pilot SustainCERT's Emissions Factor Tracking Platform to enable co-financing and co-claiming of credits and prevent double-counting (Status: In progress)

Program Development

- Productize the ESMC protocol into the MRV as a “hub” for streamlining and automating the generation of ecosystem service outcomes from agriculture at scale (Status: Complete)
- Digitize and streamline programmatic materials and functions (Status: Continuously in progress as part of core programming.)
- Integrate S3S with Marketplace platform (Status: Complete)
- Develop improved partner and member Market Platform interfaces and Salesforce tools (Status: Continuously in progress as part of core programming)
- Integrate research results from soil sampling protocol evaluation, scope 3 tool evaluation, tech assessment recommendations, water quality tool evaluation into program processes (Status: Complete.)
- Test scaled back Scope 3 Program based on improved automation and reduced data and soil sample collection (Status: Ongoing)
- Establish proxy data sources and methodologies for baseline scenario determination (Status: Complete)
- Continue development of Scope 1 WQL credit programs in priority areas (Status: On pause due to uncertain buyer interest)
- Implement Scope 3 WQL approach and evaluate business case and pricing options (Status: Complete)
- Incorporate expansion of WQN quantification metrics and data collection approach (Status: Complete)
- Develop Biodiversity program and test in Missouri Project (Status: Year 1 complete. Year 2 in progress.)
- Establish formal credit sales/transaction process and contracts (Status: Complete)

Monitoring, Reporting and Verification (MRV) platform and Gridded Land Ledger

- Build Monitoring, Reporting, and Verification (MRV) platform to support producer enrollment, data collection, and outcomes quantification for pilot and market projects (Status: Complete)
- Refine MRV data requirements and automate MRV processes to store and support data flow between different stages in the product lifecycle including DNDC, SustainCERT's S3S, and ESMC marketplace (Status: Continuously in progress as part of core programming)
- Update data collection process in the MRV to map to DNDC API requirements for modeling (Status: Complete)
- Integration of SoilStack and Stratification App into MRV (Status: Complete)
- Deploy QA/QC platform to support ESMC quality management system (Status: Complete)
- Develop Land Ledger/Marketplace Platform (Status: In progress)
- Develop Project Management Tool for data and program analysis (Status: Complete)

Deployment and Demonstration Projects

- Recruit and enroll producers to meet annual target acreage goals (Status: Complete)
- Complete Demonstration site data collection, gap filling, verification, outcomes quantification (including model uncertainty and sensitivity outcomes), asset generation, asset sales and producer payments for all project years (Status: Complete)
- Test credit generation and sale transaction processes including buyer agreement, reporting certificates, payment processing, customer account tracking (Status: Complete for 2020 and 2021 projects, in progress for 2022 projects.)
- Complete Demonstration site reports including details on asset valuation, transactions, and producer payments for all project years (Status: Complete for 2020 projects, in progress for 2021 and 2022 projects)
- Test Scope 3 scaled back data collection and processing approaches (Status: Ongoing)
- Test all aspects of program deployment to identify process efficiencies (Status: Ongoing)
- Evaluate soil stratification and carbon sampling results, uncertainty levels, and cost comparisons (Status: Continuously in progress)
- Complete Enrollment Specialist Training sessions using improved online training manual and materials (Status: Complete for 2020, 2021, 2022, and 2023 projects)
- Achieve third party validation and verification of 2020 pilots, quantification, and asset generation (Status: Complete)
- Initiate surveys and interviews to gather feedback on enrollment and program processes (Status: Complete)

Market/Business Development

- Develop financial models and refine multi-year projections to ensure financial self-sufficiency by 2025 (Status: Complete)
- Evaluate programmatic process efficiencies and cost reductions (Status: Complete)
- Award projects from General Mills Roadmap RFP to scale regenerative agriculture to 130,000 through ESMC Market Program (Status: Complete)
- Scale Eco-Harvest market program with key buyer multi-year roadmaps (Status: Ongoing)

Membership and Outreach

- Create general slide deck for member use (Status: Complete)
- Develop training program materials (Status: Complete)
- Highlight program updates to members and stakeholders through bi-weekly newsletter and mailings, social media updates and discussions with reporters (Status: Complete)
- Develop target member recruitment lists and track conversion (Status: Complete)
- Draft and distribute ESMC Annual Reports (Status: Complete)
- Develop and implement Market Program Launch Communications Plan, budget and timeline (Status: Complete)
- Develop and implement post-launch Communications Plan including press releases, updates on Eco-Harvest projects, updates to eligibility documents, and corresponding website updates (Status: Complete)
- Develop Communications Strategy and Branding Plan (Status: Complete)
- Develop online Knowledge Portal and training hub in connection with Producer Portal (Status: Ongoing)
- Develop new membership tier for large CPG and agribusiness companies (Status: Complete)
- Recruit and engage potential buyer members to support projects in development (Status: Complete)

2.5. Have any of the major goals/specific aims or milestones for the project changed since the award? If so, please list the goal(s) that have changed and provide justification for the change from the approved goals. (up to 207-word limit; current word count = 1)

Yes, Objective 5 (Quantify technical capacity for carbon and water quality outcomes based on soil carbon sequestration potentials by soil types, land management approaches and climate) was not completed due to a technical working group recommendation. Lengthy discussions with Soil Carbon Research working group members, Science Advisors, and technical partners concluded that, based on current understanding of the impacts of key practice changes (i.e., cover crops and tillage reduction) on soil carbon sequestration, reaching the technical soil carbon storage capacity of soils will likely not be achieved for possibly several hundred years. Further, unpredictable environmental characteristics resulting from climate change are certain to influence soil carbon storage potential, which complicates any long-term, land management decision-making that might result from a contemporary determination of the technical capacity of various soil types to store carbon. Technical working group members suggested exploring instead the carbon storage potential across soil types in the short-term (5-10 years) but recommended focusing resources first on key questions related to soil sampling, including stratification, sampling density, the role of remote-sensing and proximal-sensing technologies, and the relationship between field-level and project-level soil organic carbon results and crediting outcomes. Accordingly, the question of short-term soil carbon storage will be revisited as part of ESMC/ESMRC's impending research priorities.

2.6. What was accomplished under the goals/specific aims or milestones for the project? Please describe in detail, 1) major activities; 2) specific objectives; 3) significant results, including major findings, developments, or conclusions (both positive and negative); and 4) key outcomes or other achievements. Include discussion of stated goals not met. The emphasis in reporting in this section should focus on accomplishments. In the response, emphasize the significance of the findings to the scientific field. This section can be as technical as the PI would like. (up to 5,000-word limit; current word count = 4,818)

ESMC/ESMRC Public Private Partnership and Community of Practice Formed

From the project inception in January 2019, ESMC successfully built a public-private partnership, collaborative network, and community of practice to achieve progress in developing an advanced ecosystem services market program for US agriculture. Over the course of the project, ESMC/ESMRC Membership reached 80+ members and funders. Members and partners span organizations across the entire agricultural value chain and supply chain and include paying and non-paying members, partners, stakeholders and funders who have actively participated in the development of the goals and objectives of the project and in building a next-generation, technologically advanced ecosystem services market program. This includes food and beverage corporations, agricultural retailers and service providers, global and national environmental and conservation NGOs, ag tech providers, as well as land grant universities and research institutions, independent scientists, and federal and multiple state and local agencies.

Our non-profit, collectively enabling environment remains unique as a program operating in the ecosystem services market space and is a key reason that ESMC/ESMRC will outlast and out-perform other competing for-profit organizations in the market space. The financial support of FFAR and the matching member contributions that led to ESMC/ESMRC's successes do not have to be repaid, and ESMC/ESMRC's nonprofit status means we can deliver credible outcomes at scale for less than others will be able. The legacy of the FFAR grant and the ESMC/ESMRC consortium is an advanced, functioning, and scalable full-service ecosystem services market program and national scale infrastructure in continued development that will lead to improved ecosystem services outcomes from US agriculture at scale.

Throughout the four years of this project, we have hosted monthly member calls, monthly technical working group and strike team meetings, regular member webinars, and several 2-day meetings per year (both in-person and virtual once the COVID pandemic limited in-person gatherings) to collaborate and engage with members and project partners and share developments and research findings. We have also engaged in weekly pilot and demonstration project calls and many additional recruitment and member-specific calls on a weekly basis. Additionally, we participate in many external stakeholder organizations and collaborations, including engagement with standards bodies and certification bodies in the ecosystem services and carbon market arenas.

ESMC's Accredited Market Launched

The Eco-Harvest Market Program was successfully launched in May 2022 at an in-person member event in Washington DC. Launch signified readiness of ESMC's Eco-Harvest market program, as achieved by ESMC members and partners in our public-private partnership. Eco-Harvest, as envisioned in the FFAR grant proposal, goals, and objectives, is an end-to-end digitized and advanced technology program and platform that generates high quality, third-party accredited and verified Scope 3 supply chain credits [current supply chain market terminology calls these emissions factors (EF) or impact units (IU)] for soil carbon removals and reduced greenhouse gas emissions. The program successfully developed this

approach to overcome past and current market program challenges and is being widely emulated by other carbon offset, inset, and Scope 3 supply chain programs seeking to learn from our work and to establish similar programs. ESMC considers this a hallmark of success, in that more programs will help to achieve greater scale of impacts in shorter timeframes – all of which is necessary to combat the visible and escalating impacts of climate change and the negative impacts to US agricultural productivity and resilience.

The scientific and technical rigor of our program, enabled by the partnerships and investments of time and due diligence in our process, is reflected in our accreditation. ESMC's Eco-Harvest market program, validated by SustainCERT's Value Chain Initiative in August 2022, is the first national-scale market program for agricultural supply chain ("Scope 3") interventions and the first to be validated and verified by SustainCERT as generating high-quality soil carbon removals and reduced GHG outcomes for use by corporations. Publication of the process and means to generate these outcomes on SustainCERT's website means our process has become a public good that can be replicated by others. This level of accreditation and certification by a globally recognized authority ensures that corporate buyers and producers can rest assured that they are participating in and purchasing credible outcomes that will not be characterized as green washing, green wishing or worse. The program generates soil carbon removals and GHG emissions reductions from agricultural projects that food and beverage companies can credibly use to report verified claims against their Net Zero GHG commitments. Additionally, due to the level of rigor in Eco-Harvest, supply chain EF/IU generated and verified through our program can be shared by others in agricultural supply chains. This allows multiple corporate partners to invest in the outcomes in a credible, verified program that ensures accuracy of corporate claims. The program also generates for corporate reporting and tracking the impacts to water quality from U.S. agriculture, and we are continuing to pilot biodiversity outcomes. Both water impacts and biodiversity impacts are increasingly demanded by corporates in the supply chain and ESG arena, and ESMC/ESMRC is unique in delivering integrated outcomes from agricultural production.

Eco-Harvest connects producers to ESMC's on-the-ground partners for technical and agronomic support. The Eco-Harvest platform overcomes past market challenges with a technologically advanced infrastructure that is scalable, replicable, and cost-effective. Additional information about accomplishments related to Eco-Harvest can be found in the Program Development section below.

Research Projects - Quantification of Impacts

- The **Advanced Quantification Technology Assessment Project** focused on the identification, assessment, testing, and development of advanced tools and technologies to more efficiently and cost effectively measure and quantify soil C and net GHG changes. Phase I (desk evaluation of direct measurement tools, remote sensing tools, proximal sensing tools, and model approaches) and Phase III (commercial and scalability assessment) were completed in February and July 2021, respectively. Phase II in-field testing of the DeepC/vis-NIR and the Veris U3 vis-NIR spectroscopy technologies was completed in December 2021. (Reports are available in Section 3.1 Information Products.) Evaluation results were presented to WGs 1 and 4 and the Soil Sampling ST in April 2022. ESMC also partnered with EarthOptics to test their sensor-based soil mapping technologies in several 2022 demonstration projects. Preliminary evaluation results suggest that sampling density for carbon and bulk density can be reduced to less than 10% of current sampling densities while still achieving comparable results. ESMC will continue to assess possible reductions to sampling density – and incorporation of soil mapping technologies – in future demonstration projects.

- The **Comparative Analysis of Remote Sensing Data Solutions Project** set out to understand if and how remotely-sensed data products derived from satellite imagery could be used to reduce Eco-Harvest MRV program cost and improve data accuracy. Five remote sensing data tools were evaluated for their ability to support practice change and field boundary eligibility, data gap-filling, and continuous monitoring for program compliance, using fields selected from 2021 ESMC pilot projects. Gap-filling potential across the tools ranged from 50% (OpTIS) to 100% (Agrograph and PlanetWatchers); cover cropping and tillage accuracy ranged from 21% (PlanetWatchers) to 53% (Agrograph) and 45% (PlanetWatchers) to 61% (OpTIS), respectively; crop type accuracy (including a comparison to the USDA Cropland Data Layer) ranged from 82% to 98%, which varied by tool depending on the crop x year scenario; and planting and harvest date accuracy ranged from 0% to 86%, varying by tool depending on the year x planting date and year x harvest date scenario. Cost per acre per year was also evaluated and ranged from \$0.01 to \$0.58, depending on the acreage. Given these results, the following determinations were made: 1) CDL data is sufficient for crop data QA/QC; 2) technicians using the MRV (with access to USDA-NASS data and supported by automation technologies) are the most cost-effective and comprehensive for gap-filling agricultural events such as planting and harvest dates; and 3) remote sensing data can be useful for determining cover cropping and tillage reduction practice change eligibility and compliance.
- **Soil Carbon Technology Testing and Training Data Sets** were developed to support calibration and improved accuracy of remote sensing technologies that predict soil organic carbon (SOC) and/or bulk density (BD). Training data consists of geolocations (latitude and longitude) and SOC and BD measurements from 0-30 cm, while testing data includes only geolocations. The data sets will be made available to tool providers via an automated testing service. Tool providers will send their SOC and BD predictions to an online calculation engine that will calculate accuracy metrics and create a map of prediction errors. Tool providers can use the error visualizations to inform refinements to their prediction algorithms. True testing data will not be shared with tool providers to enable repeated testing. The data sets were completed in December 2022 and will provide ESMC with a cost-effective way of evaluating the feasibility of incorporating alternative soil sampling technologies into market and demonstration projects.
- The **Water Quality Modeling Improvement Project** to improve APEX modeling for Scopes 1 and 3 resulted in the addition to APEX of four conservation practice modules for tile drainage (Development of APEX Modules for Simulating Agricultural Conservation Practices Report). Source code was modified to provide modeling capacity for drainage water management, bioreactors, constructed wetlands, and saturated buffers. These enhancements are publicly available to anyone who uses APEX to identify best management practices, find sources of pollutants that affect water quality, and manage water resources on croplands. In August 2022, it was decided to pause the automation of APEX due to the substantial cost involved (approximately \$20-25K for 10 fields), the ongoing need for field-level simulation by a modeling professional in order to utilize the model, and limited certainty regarding buyer's needs and preferences for water credits. Resources for APEX automation were reallocated toward development of a database (derived from the NRCS gNATSGO soil database) that can be used to gap-fill select "null" values in the MRV platform.
- The **Alternative Scope 3 Water Quality Quantification Tool Evaluation Project**, completed in October 2021 (Technical Evaluation of Alternative Scope 3 Water Quality Quantification Tools

Report), resulted in the development of a customized approach for Scope 3 water quality credit generation. The ESMC Water Quality Calculator provides field-level water quality outcomes derived from 1) estimates of the baseline water quality loading for sediment, phosphorus, and nitrogen using median, eco-region-specific cropland loading rates provided by White et al. 2015 and 2) pollutant removal efficiencies derived from US EPA' STEPL model applied to baseline water quality loading to determine load reductions for implementation of eligible practices. Initial steps to embed the calculator into the MRV commenced in January 2023, with completion expected by March. The simplified quantification approach will deliver directionally accurate estimates for sediment, phosphorus, and nitrogen in a cost-effective and scalable manner relative to APEX and will be tested with data from several 2022 demonstration projects upon incorporation into the MRV.

- The **Water Quantity Asset Development Project** resulted in the development of a new Scope 3 water quantity (WQN) credit for the ESMC protocol based on agricultural management practices that are currently being quantified for Scope 3 water quality (WQL) credits. The Runoff Reduction credit uses the Curve Number method to estimate runoff quantities based on land cover, use, soil, and slope and accounts for temporal changes in precipitation and soil water content. Calculation of the Runoff Reduction credit was translated in the ESMC Water Quantity Calculator, which is currently being embedded into the MRV to automate outcomes quantification. The Water Quality Calculator will be tested with data from several 2022 demonstration projects upon incorporation into the MRV (expected by March 2023).
- The **N₂O Quantification Project** utilized measurements at long-term monitoring sites, remote sensing, neural networks, and machine learning to test and improve alternative methodologies for predicting N₂O emissions at a field scale to improve the scope and accuracy of GHG asset quantification and inform water quality modeling approaches. Preliminary results were provided by Colorado State University in December 2021 and by Michigan State University in July 2022. Final results furnished by MSU in November 2022 (Novel Surrogate Emissions for Estimating GHG Emissions and Ecosystem Services through Machine Learning and Remote Sensing Imagery) determined that regression-based machine learning models can robustly predict N₂O emissions in common row crop rotations under differing management practices, particularly when used in a hybrid model framework, where the machine learning algorithm is integrated with a process-based model and remote-sensing data inputs. Additional research is needed to evaluate predictions of N₂O emissions at different temporal and spatial sites before possible incorporation of the prediction approach into ESMC quantification processes. The goal of identifying surrogate measurements for estimating field-level N₂O emissions was unmet due to Colorado State University's failure to complete their contract agreement.

Research Projects - Monitoring, Reporting and Verification Technologies

- The ESMC **Monitoring, Reporting, and Verification (MRV) Platform** was designed to collect, validate, and process data for typical crop production scenarios. The MRV streamlines producer data collection and automates QA/QC procedures and gap-filling to generate data for modeling, with the goal of reducing data entry burden and errors. Foundational processes include: 1) collect accurate data for the parameters determined by Regrow's DNDC input and any program or certification requirements; 2) validate the accuracy and eligibility of field boundaries, specific data points entered, evidence of a practice change, and overall logic and timing of agronomic events based on a defined reporting year for a specific crop being produced; 3) implement

creation of comparison scenarios of practice change vs baseline based on known standards, quantification protocols, and agricultural science that are specific to the crop being produced; and 4) send comparison scenarios in a format compatible with the DNDC input constraints/requirements so that valid outcomes can be generated. The MRV productizes the ESMC Protocol and includes core functionality such as a landing page with programmatic information, MRV enrollment, and supporting documentation for program participation; administrative functions for managing and tracking projects; field identification and validation that includes multiple visual data layers created from external sources such as CDL and PADUS; integration of AdobeSign for completing Producer Agreements; data collection portal for gathering the data needed for modeling; database for tracking producers, fields, projects, management data, modeling outcomes, and credits generated; pre-modeling data processing modules for data validation, baseline determination, and preparation of data to transfer to modeling tools; comprehensive QA/QC steps to ensure data quality and enable third-party validation of data and processing steps; and post-modeling data processing modules for outcome calculations (including the application of uncertainty adjustments at the field- and project-levels), tracking, reporting, and preparing to transfer to the S3S tool. The MRV represents a first-of-its-kind, defensible system of data collection and processing with capabilities to evolve with a dynamic market program.

- The **Stratification App** was developed to generate highly accurate soil carbon sampling and monitoring plans for producers participating in the ESMC program. The tool collects remote sensing and digital soil mapping data which is clipped to a target field and processed. k-means clustering and Conditioned Latin Hypercube Sampling (cLHS) algorithms are used to determine soil strata and optimally locate soil sample locations to best represent potential variability in soil carbon content across a field. The Stratification App has been utilized in nearly all demonstration projects over the past three years, and ESMC's investments in this open-source tool have enabled significant performance enhancements that are available to other users. A self-hosted instance of the Stratification App has been integrated into the MRV to enable automated and scalable stratification and selection of soil sampling locations. Documentation and research reports associated with the Stratification App can be found in Section 3.1 Information Products.
- The **Project Management Portal** was developed to facilitate project development, implementation, and management through each step from project initiation through impact unit reporting. The project management portal connects directly to data stored in the MRV to provide real-time enrollment and project status information for ESMC staff, contributing to programmatic process efficiencies.
- The **Quick Stats and Data Analysis Portal** was built to enable quick high-level evaluations of project enrollment, details, and outcomes and well as deeper analysis of program data. The Quick Stats Portal includes a business intelligence data visualization dashboard that links with the MRV database and provides customized data views for producers, projects, and ESMC users. The Data Analysis Portal can be used for more in-depth data analyses through a Jupyter hub data analysis portal that will ultimately include a role-based method to access data from the MRV database. Insights generated through the Data Analysis Portal will be used to inform improvements and modifications to the ESMC program and producer participation experience.

- The **Biodiversity App** was developed as a cost-effective method to monitor and verify biodiversity observations on agricultural lands. The app includes a growing database of plant, animal, and insect species of concern and is being beta tested in ESMC's biodiversity pilots in Missouri and Kansas, where participating producers are planting vegetative buffers with native grass and flower species to support pollinator habitat and other ecosystem services.

Research Projects - Ecosystem Services Marketplace Platform and Land Ledger

- ESMC and SustainCERT collaborated to jointly develop the business and use case and minimum viable product (MVP) for SustainCERT's Scope 3 Software (S3S), which is not a function or role that ESMC/ESMRC could take on, but which is a necessary function for ESMC's fully functioning and accurate supply chain market program to operate. In the absence of the collaboration, SustainCERT would not have made the same progress for another 18 months, and ESMC could not transact sales or launch its buyer co-investment model. SustainCERT'S S3S platform uses cutting-edge LCA databases to accurately transform ESMC's ground-truthed and modeled data from interventions (i.e., ESMC projects with a market-supported practice change) into impact units and emission factors that allow corporates to account for commodity-specific stock changes in their GHG inventories.
- ESMC contracted with SustainCERT and Heartland to connect the Eco-Harvest MRV platform to SustainCERT's Scope 3 Software (S3S) to track generated credits, prevent double counting, and enable co-investing by multiple supply chain buyers. Requirements for this integration, which will streamline and digitize the verification process for scale, are currently under development by SustainCERT. Meanwhile, Heartland has begun to develop the necessary components to enable the Marketplace platform and outcomes tracking to facilitate the buying and selling of impact units. Once these requirements are defined, the MRV platform will integrate closely with the S3S and will leverage additional tracking and payment systems such as Salesforce and Bill.com.
- A digital verification system (dMRV) of the ESMC program was developed to expediate third-party verification of salable credits, incorporate automated and human-driven QA/QC and program management controls, meet corporate reporting schedules, and reduce verification costs. Historically, verification of agricultural projects typically accounted for 75% of the credit costs; ESMC's vision of a fully digitized verification process was embraced by global certification and verification bodies Gold Standard and SustainCERT.
- ESMC is building a white-label Eco-Harvest marketplace to facilitate the transaction of impact units. This web application and collection of APIs will connect supply and demand for verified Scope 3 GHG Impact Units (IUs) for corporate reporting within ESMC's program regions. The marketplace will function as an integrated match-making service to sell crop-specific Impact Units. The marketplace leverages (a) ESMC's member network of key stakeholders, (b) set rules and processes for Impact Unit generation and transfer, and (c) customized user accounts to facilitate both co-investment and co-claiming across value system impact layers by cropping system and geography.

Research Projects - Quantifying and Achieving Ecosystem Service Capacities

- The **Scope 3 Soil Sampling Protocol Evaluation Project** evaluated stratification, soil carbon sampling protocols, and measurement results for 2020-2021 demonstration projects. RES and Regrow evaluations ([RES IL](#), [RES GM](#), and [Regrow](#) reports) showed that significantly lower soil

sampling density on a field- or project-level basis has virtually no impact on soil carbon and crediting outcomes but substantially lowers costs. As a result, ESMC modified its soil sampling protocol to a sampling density of one sample per four acres (versus 3) for 2022 projects. Stratification and soil sampling density remain critical research areas for ESMC from the perspectives of cost, risk, and relationship to accurate and defensible outcomes and will continue to be explored in future programming.

Model and Tool Development

- **Model development to increase scientific rigor and meet external validation and verification requirements** included refinement of calibration and validation data for DNDC and APEX models. Investigations to create regional versions of DNDC with specific calibrations demonstrated no greater accuracy than a national version. Detailed literature reviews to establish regional DNDC validation did not identify enough long-term studies to establish validation domains for individual program regions, practice changes, or production systems. A single validation domain was proposed to estimate structural model uncertainty for DNDC for SOC and direct N₂O emissions across multiple ESMC Program Regions, cropping systems, and practice categories and was utilized in the quantification of outcomes for 2021 projects ([DNDC Validation Process for ESMC 2021 Pilots](#)).
- **Improvements to model efficiency and data transfer** included scalable architecture for model simulations and data processing; a harmonized APEX model input data approach; automated validation and gap-filling processes and export from MRV v3 to DNDC and APEX; and API connectivity to DNDC to automate efficient data transfers.
- **DNDC model preparation to quantify credits** included work to evaluate structural model uncertainty to improve data feeds to the S3S platform. The functionality to generate separate uncertainty estimates for baseline and project scenarios was developed and integrated into the post-modeling data processing modules for outcome calculations noted in the MRV outcomes section above.
- **Regrow's OptIS product development** added algorithms for extended areas and temporal updates through 2020. OptIS was used in gap-filling, QA/QC, and data corroboration for 2020 projects and in 2021 projects will confirm practice changes and inform historical practices when historical data from producers is missing.

Protocol Development

- ESMC's program outcomes from 2020 and protocol were certified by SustainCERT/Gold Standard and determined to be in alignment with the Value Change Initiative. This approval of our program will streamline verification of outcomes in 2021 (currently underway), and subsequent years going forward.
- The ESMC protocol was updated (v10, to be submitted in early 2023) to reflect recent major changes to the ESMC Program:
 - Outlines the process to quantify carbon and water quality outcomes for combinations of geography, production system, and practice type
 - Updates our uncertainty process to meet 2020 certification and SustainCERT S3S integration
 - Adjusts the ESMC data flow to generate emissions factors for corporate reporting in addition to credits via S3S integration

- Describes the process to generate cost-effective water quantity and water quality outcomes through a simplified approach that is appropriate for current corporate reporting standards
- A program guide was developed for internal use and details the protocol and other technical considerations of the end-to-end process to generate environmental attributes in the ESMC program including:
 - Updated data requirements for DNDC modeling and for S3S and updates to the baseline and project definition process in the ESMC MRV to generate emissions factors
 - Characteristics of the market program and differentiation between "market" and research" as it relates to the verification of ESMC outcomes
 - Quality Management Plan and QA/QC functionality in the ESMC MRV to improve data quality and streamline verification via the SustainCERT dMRV program
- ESMC developed a [program summary](#) for members to review to understand the capabilities of the ESMC program following the ESMC market launch.
 - ESMC continues to advocate standard setting bodies including GHGP, SBTi/SBTN, and VCI for a system that works for both producers and ESMC members that is also aligned with global climate targets
 - Efforts will proceed to scale and expedite the verification of ESMC Scope 3 GHG outcomes with the continued assistance of the SustainCERT dMRV platform

Certification and Standards

- Validation and verification for 2020 cropland projects from the Midwest Corn and Soy, Great Lakes, and Southern Great Plains regions for Gold Standard Value Change Initiative (VCI) certification was completed, and certification by Gold Standard and ESMC was announced in August 2022. Certification to the VCI standard provides assurances to buyers that the GHG reductions and removals generated through ESMC's market program are credible, science-based, and eligible for use in corporate Scope 3 GHG inventory reporting. The VCI standard is aligned with the GHG Protocol's Scope 3 Reporting Standard, the relevant international guidance for voluntary GHG inventory accounting.
- ESMC's engagement in the GHG Protocol's (GHGP) continued development of the Land Sector and Removals Guidance includes technical working group, pilot project and reviewer group participation. ESMC drafted white paper recommendations for GHGP review and continues to advocate for Scope 3 market programs considerations within the GHGP standard.
- ESMC is a member of the corporate engagement program of the Science Based Targets Network (SBTN) which set standards for corporate accounting and target setting for water quality, biodiversity, and other environmental metrics. ESMC provided feedback to SBTN freshwater guidance released in May 2022 to ensure lessons learned are considered during continued evolution of SBTN guidance.

Program Development

ESMC/ESMRC launched the Eco-Harvest market program in May 2022. Launch signified readiness of ESMC's Eco-Harvest market program, an end-to-end digitized and advanced technology program and platform to generate high quality, third-party verified scope 3 credits for soil carbon removals, reduced greenhouse gases, and water impacts from U.S. agriculture. Eco-Harvest establishes clear pathways of engagement in ecosystem service market and connects producers to ESMC's on-the-ground partners for

technical and agronomic support. The Eco-Harvest platform overcomes past market challenges with a technologically advanced infrastructure that is scalable, replicable, and cost-effective. Program documentation was created to outline market program guidelines, eligibility requirements, the enrollment process, and eligible program regions. Eco-Harvest market operations cover the Northern and Southern Great Plains, the Midwest Corn and Soy, and Great Lakes Regions, and include corn, soy, wheat, and alfalfa production systems and cover cropping, reduced or no-till, and nutrient management practice changes. Additional program regions, crops, and practice changes will continue to be piloted and incorporated into the ESMC market program as market readiness criteria are achieved.

Deployment and Demonstration Projects

ESMC successfully launched 19 deployment and demonstration projects covering 104,081 acres and 198 producers across the US. Over the course of this project, ESMC Project Managers hosted regular calls with pilot partners to identify, discuss, and make progress on demonstration project objectives and milestones. Demonstration projects enabled ESMC to test and refine approaches to producer enrollment; soil sampling; deployment of eligible practices; baseline determination; data collection, gap-filling, and QA/QC; outcomes quantification; credit generation and sale; and producer payments and to identify process efficiencies.

The Eco-Harvest Market Overview for Producers describes the eligibility criteria for participating in ESMC's Eco-Harvest Program. Eco-Harvest is currently operable in the Northern Great Plains, Southern Great Plains, Great Lakes, and Midwest Soy and Corn Belt regions and supports corn, soybean, wheat, alfalfa, and oat cropping systems. Producers implementing cover cropping, tillage reduction, and nutrient management practice changes are eligible for enrollment. Work is underway to expand Eco-Harvest coverage to additional regions, crops, and practice changes.

Project Reports presented to project partners for 2020 and 2021 projects detailed asset valuation, transactions, and producer payments. Example reports describing pilot outcomes and credits and emission factors generated include: Sustainable Northwest Project, ADM IL Corn 2020 Project Summary, and VCI Project MN TNC.

ESMC enlisted the support of University of Maine, University of Nebraska-Lincoln, and American Farmland Trust to investigate producers' and partner organizations' experiences with participating in the ESMC payment-for-ecosystem-service-program. Working with ESMC staff, the research group developed survey instruments and focus group prompts to gather insight into the perceptions of project coordinators, enrollment specialists, and participating producers regarding the ESMC program design, enrollment experience, and other programmatic functions and to learn about the factors driving producers to adopt soil health practices. Data collection was stymied by a lack of interest in participation across all stakeholder groups, resulting in just one focus group held out of seven planned and no survey responses. Focus group participants, which included project coordinators and enrollment specialists, agreed that heavy reliance on partner organization staff time to execute projects poses a barrier to scaling, as do data collection and entry and soil sampling. In place of the survey, one-on-one interviews were held with three producers representing two different pilot projects who were open to sharing their feedback. Producer interviewees were aligned in their opinion that data entry was the largest challenge to their participation, followed by lack of clarity regarding practice change eligibility. Producers also concurred that outcomes payments in the range of \$10-20 per acre may not be sufficient to incentivize

participation given the data burden and lengthy contracts for use of specific practices. Despite the perceived challenges, research participants expressed optimism for ESMC's role in delivering compensation to farmers for the provision of ecosystem services.

2.7. Discuss efforts taken to ensure the approach is scientifically rigorous. Include approaches taken to ensure robust and unbiased results.

All research projects were developed under the guidance of Technical Working Groups and a committee of Science Advisors with experience and expertise in key areas including greenhouse gas emissions, soil carbon, water quality and water quantity, data sciences, and technology development. Project milestones were shared with the Working Groups during monthly meetings, or as they became available, and input was solicited to ensure alignment with scientific principles and consistency with current research. Model and tool development was informed by relevant peer-reviewed literature, and sources were referenced in the associated technical documentation for transparency.

Quantification of outcomes for ecosystem services is still an evolving science. Our members and partners have recommended multiple approaches for calibrating and validating process models used for outcomes quantification and how to calculate uncertainty. We have navigated these recommendations by seeking input from a range of scientific experts and then documenting the decisions accordingly. A Quantification Outcomes Advisory Committee was assembled to review these decisions and reaffirm adherence to scientific principles and alignment with the current state of the science for ecosystem services quantification outcomes. As ESMC continues to refine quantification methodologies for carbon, GHG, and water outcomes, technical partners are required to provide updated documentation which undergoes both internal and expert review prior to implementation within the ESMC program. Further, ESMC has implemented several approaches within the MRV to support robust, scientifically-defensible outcomes, including pre-modeling data processing modules for data validation, baseline determination, and preparation of data to transfer to modeling tools; comprehensive QA/QC steps to ensure data quality and enable third-party validation of data and processing steps; and post-modeling data processing modules for outcome calculations, including the application of uncertainty adjustments at the field- and project-levels.

2.8. List key stakeholders that could be served by results of this project.

Producers, enrollment specialists, corporate buyers, researchers, investors, society

2.9. How have the results of this reporting period been disseminated to communities of interest? Describe how the results have been disseminated to communities of interest. Include any outreach activities that have been undertaken to reach members of communities who are not usually aware of such activities, to enhance public understanding and increase interest in learning and careers in science, technology, and the humanities. Reporting the routine dissemination of information (e.g., websites, press releases) is not required. For awards not designed to disseminate information to the public or conduct similar outreach activities, a response is not required; the grantee should write “nothing to report.” A detailed response is only required for awards or award components that are designed to disseminate information to the public or conduct similar outreach activities. Note that scientific publications and sharing research sources will be reported under Information Products. (up to 1,000-word limit; current word count = 965)

Dissemination of information, outcomes, and learnings to increase Member understanding of project goals and objectives, to solicit feedback, and to increase Member engagement in initiating and guiding research, development and deployment activities was accomplished through regular Member and WG calls and meetings, topical webinars, Basecamp postings, and retail calls and meetings with members. Information was shared with the public and interested stakeholders through newsletter articles, conference presentations, ESMC’s website and social media accounts, press releases, and individual staff responses to inquiries received through the website email feature. ESMC was also highlighted in numerous articles in popular media throughout the course of the project.

Events and postings that occurred throughout the course of the project include:

ESMC Website and Branding:

- Created ESMC logo, presentation, and document template
- Developed and launched organization website with interactive capabilities such as market and pilot project map, social media postings, and program FAQs
- Created an internal branding plan (for staff use) and external branding plan (for members and the public) to ensure ESMC’s brand and language are correct and consistent; both internal and external branding plans are available if desired
- Developed and implemented 6-month Communications Plans to coordinate website launch, graphics development and logo release
- Developed ESMC-specific graphics that include over 35 graphic icons, a market program graphic, an ESMC timeline (highlighting stages in ESMC’s growth and development), as well as pilot-specific graphics such as timelines. All graphics are available if desired.
- Created Eco-Harvest program name, tagline, talking points and logo and attained copyright protection
- Developed Eco-Harvest Branding Plan and Social Media toolkit and launched Eco-Harvest section of website. Drafted and printed Eco-Harvest launch materials.
- Established social media presence through collaborations with partners to amplify events and news. Increased frequency of postings and use of photos and graphics. Social media statistics as of September 2022 include:
 - Twitter: 653 followers
 - Facebook: 141 followers

- LinkedIn: 1385 followers
- Newsletter subscription list: 1999 subscribers

Member Meetings:

- Monthly member calls
- Spring Member Meeting held May 11-12, 2020
- Summer Member Meeting held August 11-12, 2020
- Winter Member Meeting held January 12-13, 2021
- Spring Member Meeting held May 19-20, 2021
- Virtual Member Meeting held January 25-26, 2022
- In person Member Meeting & Eco-Harvest Launch held May 17, 2022 (Washington, DC)

ESMC/ESMRC Member Webinars:

- April 8, 2020: ESMC Protocol Update
- April 13, 2020: Gold Standard-SustainCERT Webinar
- April 27, 2020: NRCS Conservation Assessment Ranking Tool (CART) Webinar
- August 6, 2020: Adaptive Multi-Paddock (AMP) Grazing Research Project Webinar
- September 25, 2020: NYC Climate Week Webinar on Farmer Economics and Climate Smart Agriculture
- October 20, 2020: ESMC Protocol Update
- November 13, 2020: ESMC Webinar on Monitoring, Reporting, Verification (MRV) Platform – Needs, Planning, and Data Portability
- November 30, 2020: ARPA-e Technology Investments Webinar
- December 9, 2020: ESMC Scope I GHG Protocol Update: Side-by-Side Comparison of ESMC, CAR, and Verra Protocols
- February 17, 2021: Advanced Soil Carbon Quantification Technology Assessment Project Phase 1 Report – Patrick Lawrence, Repliculture
- February 23, 2021: Mississippi River Basin State Policy Report Presentation – Heidi Peterson, Sand County Foundation
- June 8, 2021: Advanced Soil Carbon Quantification Technology Assessment Project Phase Report and Project Wrap-Up – Mike Komp, CTIC; Patrick Lawrence, Repliculture
- June 17, 2021: Integrating Sensors, Remote Sensing, and the DNDC Model for Quantifying GHG Emissions: An ARPA-e SMARTFARM Project
- August 26, 2021: DNDC Update: Calibration, Validation, and Preliminary Uncertainty Results — Bill Salas, Regrow
- November 13, 2021: Alternative Scope 3 Water Quality Quantification Tool Evaluation: Final Report and Recommendations – Michelle Schmidt, Tetra Tech
- July 12, 2022: GHG 101 – Grant Iverson-Lane, ESMC

Press Releases

- March 17, 2022: ESMC and Benson Hill Partner to Support Midwest Farmers Participation in Agricultural Carbon Markets
- March 22, 2022: Partnership for credible reporting on carbon intensity of agriculture commodities
- May 3, 2022: ESMC Partners with the Conservation Innovation Fund to Create Innovative Water, Carbon, and Biodiversity Credit Opportunities in the Mid-Atlantic

- May 10, 2022: ESMC and The United Sorghum Checkoff Program (USCP) Launch Project for Sorghum Farmers to Create Ecosystem Services Credits
- May 17, 2022: FFAR-Funded ESMC Launches Eco-Harvest, an Ecosystem Services Market Program
- June 7, 2022: General Mills Invests \$3 Million to Scale Eco-Harvest by Ecosystem Services Market Consortium
- June 21, 2022: Ecosystem Services Market Consortium Publishes 2021 Annual Report Showcasing Innovation, Growth and Collaboration
- August 23, 2022: ESMC is the First US Market Program to Achieve Multiple Milestones
- September 14, 2022: ESMC is a Partner in Five USDA Climate-Smart Commodity Grant Awards
- September 16, 2022: Mid-Atlantic Dairy Farmers Win in USDA Climate-Smart Commodities Announcement
- November 8, 2022: ESMC and SLM Partners Launch Eco-Harvest Pilot Project Focusing on Organic Systems to Create Ecosystem Services Credits
- November 17, 2022: ESMC and the Almond Board of California Launch First Project for Almond Growers to Create Ecosystem Services Credits
- December 15, 2022: Collaborators Launch Eco-Harvest Project for Southern Cotton Farmers to Participate in Ecosystem Services Markets

A comprehensive list of press releases issued by ESMC over the course of the project is available at <https://ecosystems-servicesmarket.org/news/esmc-news/>

ESMC/ESMRC Trainings:

- Developed presentations and materials for ESMC Enrollment Specialist Training including easy-to-follow graphics and video tutorials
- Conducted numerous Pilot Enrollment and data entry trainings for producers and pilot partners
- Held numerous Soil Sampling and Stratification Trainings for pilot partners and soil samplers
- Developed layout for Knowledge Portal requirements for Enrollment Specialist Training including standard training materials, slides and talking points
- February 10, 2021: ESMC Enrollment Specialist Training at National Association of Conservation Districts Annual Meeting
- March 24, 2021: ESMC Enrollment Specialist Training at American Society of Agronomy
- July 28, 2021: ESMC Enrollment Specialist Training at Soil and Water Conservation Society Annual Meeting
- January 21, 2022: 2021 Pilot Partner review meeting to outline recommendations for 2022 partner training
- Presented two case studies on Enrollment Specialist Training at SWCS 2022 Conference

ESMC/ESMRC External Presentations

- ESMC presented at over 100 external fora during the grant period. A full list is available if desired.

2.10. Describe challenges or delays encountered during project and actions that were taken to resolve them. Only describe significant challenges that may have impeded the research and emphasize their resolution. (up to 500 word limit; current word count =497)

Multi-sector coalition building and engagement: Member recruitment is a resource-intensive undertaking, particularly because the roles and needs of our partners – who span the agricultural value chain – along with their knowledge and experience around Scope 3 supply chain operations, accounting and reporting, verification and market standards, etc., are highly variable and require considerable effort to differentially meet. This ongoing effort is separate from the other needs of developing a market, launching research and demonstration projects, developing infrastructure, organizing and leading technical working groups, and incorporating learnings and findings into our program.

Competition in our pre-competitive forum: Despite being a pre-competitive forum and members signing agreements to protect against conflicts of interest, we experienced members duplicating our programming and exiting to launch their own for-profit efforts. While this was not preventable, we do not have the resources as a non-profit to compete for producers, for media attention, and to launch in comparable time-frames as other organizations. Although we were not first to market, we are confident we will be best to market once we are operating at scale and that we can outperform these actors at lower cost.

COVID pandemic: The pandemic prevented our members from gathering in person, hampering relationship-building, collaboration, communications, and program growth. COVID also impacted the rollout and scale of demonstration projects due to the inability to travel and meet with producers. Our Producer Circle was unable to meet in person to learn more about the program, provide input, and become effective program ambassadors. The ability to complete soil sampling and analysis in a timely manner was also challenged. To overcome these obstacles, we converted to on-line meetings and 1:1 retail work with individual members and worked with alternative soil sampling organizations and labs where possible. In spite of this effort, we found that many members were still not fully up to speed on our successes and achievements; however, project partners were generally understanding about sampling and analysis delays given the circumstances.

Soil carbon quantification: Soil carbon sampling is resource-intensive and expensive, and there remains debate about how much soil sampling is required for accurate baseline determination and tracking of temporal changes in soil carbon. Sampling in our first year of demonstration projects – based on the protocol designed by our cooperating scientists – cost \$12/acre, which is untenable for buyers who are generally seeking sampling costs closer to \$1/acre. While we have reduced costs considerably due to investments in stratification and partnerships with technology companies, soil sampling remains the single highest cost of program operations and the greatest source of risk due to human error, tool failure, lab analysis delays, and the whims of nature. We continue to work with soil sampling and analysis partners to ensure compliance with our protocols, and we have developed training and testing data sets that require tool developers to demonstrate the accuracy of their tools. We also continuously evaluate alternative sampling technologies for accuracy, repeatability, and commercial viability – all of which are crucial to reducing costs and scaling our program.

2.11. What opportunities for training and professional development has the project provided during this reporting period? If the research is not intended to provide training and professional development during this period, state “Nothing to Report.” For all projects reporting graduate

student and/or post-doctoral participants, grantees are encouraged to describe how Individual Development Plans (IDPs) are used to help manage the training for those individuals. (up to 500-word limit; current word count = 3).

Nothing to Report.

2.12. Please indicate the number of undergraduate and graduate students, post- doctoral scholars, or other educational components involved during this reporting period. If other education components are involved, please describe them in detail. (up to 500-word limit; current word count = 57)

None. Efforts over the grant period to secure graduate, undergraduate or post docs with soil science and/or water quality expertise were not successful via posted positions and socializing of the positions among members, collaborators, and partners. A direct attempt to secure research soil scientists under contract with OSU was unsuccessful due to impacts from the COVID pandemic that resulted in prolonged contract negotiations between ESMC and OSU that were never finalized. Besides staff turnover occurring at the office and in the department we were engaged with at OSU, contracting was made more difficult due to the University's legal team being understaffed due to COVID pandemic effects on their operations. Accordingly, funding for these positions was reallocated in support of other grant objectives.

3. Information Products

3.1 Please list the type(s) of information products (e.g., scholarly publications, reports or monographs, workshop summaries or conference proceedings, video, audio, images, models, software, curricula, instruments or equipment, intervention, etc.) produced during this reporting period resulting directly from the FFAR award.

Research

The Advanced Soil Quantification Technology Assessment Project Report serves as a resource for incorporating advanced tools and technologies to more efficiently and cost effectively measure and quantify soil C and net GHG changes into ESMC's soil sampling protocol. Reports are available here:

- [Tech Assessment Phase I: Remote Sensing and GIS Report](#)
- [Tech Assessment Phase I: Direct Measurement and Proximal Sensing Report](#)
- [Tech Assessment Phase I: Modeling and Stratification Report](#)
- [Tech Assessment Phase I: Final Report](#)
- [Tech Assessment Phase II: Final Report on Visible Near-infrared Penetrometer Testing](#)
- [Tech Assessment Phase II: Measuring Soil Carbon with Intensive Soil Sampling and Proximal Profile Sensing](#)
- [Tech Assessment Phase III: Final Report](#)

[Comparative Analysis of Remote Sensing Data Solutions for Scaling Ecosystem Services Markets](#) – Technical assessment of remote sensing data products from several third-party providers, including comparison of pricing, feature set, and accuracy of predictions relative to producer data recorded in the ESMC Eco-Harvest MRV platform.

Novel Surrogate Emissions for Estimating GHG Emissions and Ecosystem Services through Machine Learning and Remote Sensing Imagery

Development of APEX Modules for Simulating Agricultural Conservation Practices Report – Describes enhancements to APEX model source code to provide modeling capacity for evaluation four additional conservation practices (drainage water management, bioreactors, constructed wetlands, and saturated buffers)

Technical Evaluation of Alternative Scope 3 Water Quality Quantification Tools Report – Technical assessment of potential water quality quantification tools for Scope 3 water quality credits

Summary of ESMC Water Quality Calculator – Outlines simplified approach for providing nitrogen, phosphorus, and sediment load reduction estimates for Scope 3 water quality credits

Summary of ESMC Water Quantity Calculator – Outlines approach for using Curve Number method to quantify benefits of conservation practices that reduce runoff from agricultural fields

Metrics for Evaluating Water Issues to Inform Prioritization of Funding Conservation Practices – Summarizes nationally available metrics that buyers can use to support the identification of potential water issues in their sourcing regions and the credit types that ESMC offers to address different types of water risk

Stratification App readme files: GM KS pilot (v0.0.1) and the 2020 IL Corn and TNC MN pilots (v0.0.2) – Describe process for optimally locating soil samples to capture variability in soil organic carbon content across a field

SoilStack User Guide – Provides guidance to soil samplers using SoilStack application for collecting and tracking soil samples

Evaluation of Ospats+ Stratification Methodology – Description of the assessment undertaken to determine the feasibility of using Ospats+ in place of the combined conditioned Latin Hyper-cube Sampling (cLHS) and k-means clustering approach used in ESMC's current stratification process

Determining Optimal Sample Size Using Power Analysis – Evaluation of an alternative power analysis method to recommend a soil sampling density that explains the variance observed in the co-variate ancillary data layers used in the ESMC soil stratification process

Field Buffer Analysis – Evaluation of the impact of field buffer size on the sampleable area and generation of soil sampling locations

DNDC Validation Process for ESMC 2021 Pilots – Describes the validation and uncertainty quantification processes developed by Regrow to enable the use of the DNDC model in the ESMC Market Program.

DNDC Validation Report: ESMC Chesapeake Bay Program Region – Example report detailing Regrow's validation of the DNDC model for specified crops within a specific production region.

Rotational Tillage White Paper – Summary of factorial simulation study using DNDC soil model to analyze the differences in GHG outcomes between multiple types of tillage practices across a range of soil and climate conditions

Biodiversity Monitoring Literature Review and Development of a Minimum Viable Product (MVP) Web Application – High-level literature review of relationship between ESMC eligible practice changes and impacts on biodiversity, landscape assessment of biodiversity monitoring applications, and overview of ESMC’s minimal viable product (MVP) progressive web application for agricultural biodiversity monitoring.

Assessment of Ecosystem Service Marketplace Programming – An investigation of producers’ and partner organizations’ experiences with participating in the ESMC payment-for-ecosystem-service-program

Protocol and Standards

ESMC’s Market Program Protocol version 9.0 was finalized in November 2021 and shared with Members via a webinar to review significant changes from prior versions. Protocol v 9.0 summary document ([ESMC Protocol Summary v9 2022](#)) and a condensed outline (ESMC Protocol v9 Overview) were developed and shared with members. Version (v10.0) of the ESMC protocol is planned for release in early 2023. Additionally, the [ESMC Protocol Guide for Eco-Harvest](#) was developed to provide a concise overview of the ESMC Market program.

Deployment and Demonstration Projects

The [Eco-Harvest Market Overview for Producers](#) describes the eligibility criteria for participating in ESMC’s Eco-Harvest Program.

The ESMC Producer Portal serves as the Program enrollment and data tracking platform and is available here: <https://www.esmcportal.org/>

The following Producer Reports describe example pilot outcomes and credits and emission factors generated:

- [Sustainable Northwest Project](#)
- [ADM IL Corn 2020 Project Summary](#)
- [VCI Project MN TNC](#)

Producer and project contracts and materials are under currently under legal review for adherence to the Ag Data Transparent principles and will be revised accordingly pending review outcomes.

Other

[Meridian Institute Report #1: Reflections on the Ecosystem Services Market Consortium 2022 Market Launch Plan](#)

[Meridian Institute Report #2: Key Insights for the Ecosystem Services Market Consortium’s Critical Path to the 2022 Launch](#)

[ESMC Market Supply Analysis for the Western Range](#)

[ESMC 2021 Annual Report](#)

[ESMC 2020 Annual Report](#)

3.2. Please provide a list of citations for the information products produced during this period

Not applicable

3.3 Are there publications or manuscripts accepted for publication in a journal or other publication (e.g., book, one-time publication and monograph) during the reporting period resulting directly from the FFAR award?

No

3.4 Website(s). List the URL for any internet site(s) that disseminates the research activities. A short description of each site should be provided. It is not necessary to include the publications already specified above

No additional details – see section 3.1 above

3.5 Have inventions, patent applications, and/or licenses resulted from the project (U.S. and international)? If yes, indicate the invention, patent application(s), and/or license(s).

No

	# Filed (Enter Numeric Value)	# Approved (Enter Numeric Value)	Patent Numbers (Separated by commas)
Patents	#	#	
Copyrights	#	#	
Trademarks	#	#	
Inventions	#	#	

3.6 Are any of the information products produced during this grant that are confidential, proprietary, or subject to special license agreements? If so, please list them below and describe why they must remain confidential. Also, note if (and when) you plan to make these data publicly available in the future or if they must remain confidential indefinitely. (up to 500-word limit)

Under agreement with ESMC/ESMRC members, and to protect their engagement and investments, ESMC/ESMRC reports are accessible to members only for twelve months upon completion to allow them time to utilize the information prior to being shared with the general public.

Per this policy, the following list of reports will be made available to the public by the dates noted:

- Novel Surrogate Emissions for Estimating GHG Emissions and Ecosystem Services through Machine Learning and Remote Sensing Imagery (December 2023)
- Neural Network for Gap-filling and Estimating N₂O Emissions (Colorado State University)
- Advancing Re-Sampling Designs to Confidently Detect Temporal Change in SOC Stocks (January 2024)
- Assessment of Ecosystem Service Marketplace Programming (December 2023)
- Comparative Analysis of Remote Sensing Data Solutions for Scaling Ecosystem Services Markets (December 2023)
- Biodiversity Monitoring Literature Review (December 2023)

3.7 Beyond depositing information products in a repository, what other activities have you undertaken to ensure that others (e.g., researchers, decision makers, and the public) can easily discover and access the listed information products? What other activities have you undertaken to ensure that others can access and re-use these data in the future?

All program materials, including documents, reports and product links are shared with members, Working Group and Strike Team members, and Ex Officio USDA, DOE ARPA-e and US EPA representatives through BaseCamp, our online member document repository. Additionally, we generally share and review products during monthly member meetings or specifically scheduled topical webinars to share the content of these materials in addition to posting them.

Publicly released program materials are posted on ESMC's website and announced and shared in our monthly Newsletter and monthly News of Note column to denote their availability to the general public. We frequently also share documents and reports as electronic handouts (sometimes, in hard copy) while participating in conferences and workshops and doing external invited presentations.

To ensure the appropriate access to and reuse of data now and in the future within the confines of producer privacy agreements, we participate in multiple research collaborations, such as NASA Harvest and collaborations with other FFAR grantees.

4. Data Management

4.1. During this period, did the project generate any data? Data generation includes transformation of existing data sets and data from existing resources (e.g., maps and imageries).

Yes

Please list the data generated in this reporting period

We collected the following field and farm scale data for deployment and demonstration projects over the course of this project:

- Crop type & historical average yield
- Animal type & historical average stocking rates
- Historical farm management practices (tillage, cover cropping, irrigation, fertilization events)

- Practice and systems change(s) undertaken, including dates, required to qualify for credit generation
- Field boundaries, soil sampling strata, sample points/coordinates, sampling dates, lab results (TOC, bulk density, P, pH), cost per sample, cost per acre
- Presence of protected areas on farms and ranches claimed by producers as within project boundaries
- Acreage and number of fields by crop type and practice change
- Modeled GHG emissions and soil carbon values from DNDC
- Modeled water quality values from APEX

4.2 If you list multiple data sets, are these data sets related? If so, please provide a short description of how they are related

Data sets listed in 4.1 are all components of on-farm/ranch demonstration projects. The modeled outcomes are related to these farm/ranch specific data sets and form the basis of outcomes generated for corporate claims.

4.3. Please provide copies of relevant metadata records to support FFAR’s mission of enhancing the discoverability of FFAR funded project data and information products. Upload copies of records and a simple file inventory, if necessary, in a compressed folder.

FFAR is developing a more nuanced Data Management Policy. As part of this effort, we are introducing new questions into the Annual Reports on Data. All questions on this page are required to have a response. In cases where your answer may be repeated, you are welcome to copy-paste relevant information. Thank you for your patience as we update our reporting.

Due to producer privacy policies and contractual agreements with producers, ESMC does not share producer data with external parties, except in stated cases when sharing for research purposes is allowed. We will continuously share summaries of outcomes at an aggregated and anonymized level once project sites have been assessed and evaluated for program purposes.

Are the data experimental, observational, simulation, or compiled/derived? Check all categories that apply.

Observational: Data captured through observation of behavior or actions of the situation as is (e.g., surveys, sensors, monitors) ✓

Simulation: Data generated by developing and applying a computer model that simulates system behavior (e.g., ecosystem, behavioral) ✓

Compiled data: Data compiled using extant data sources and refined to develop a meta-analysis or systems review

For each checked box, please provide a brief description of the data including where the data are collected / sourced.

Deployment and demonstration project data is stored on a secured server in a SQL relational database that is part of ESMC/ESMRC's digitized platform. Producer identifying information is anonymized when it is exported outside of the digitized platform (e.g., for modeling). Access to the database by authorized users requires a VPN connection and multifactor authentication. Observational and simulation data stored in the database includes:

- User input data such as field boundaries and agricultural management history collected via the ESMC MRV User Interface. Data are provided either by producers, their advisors, or a project administrator.
- Supplementary field-level data from reputable external sources used for data validation and gap-filling
- Normalized soil data collected by soil samplers and analyzed in the lab
- Modeled carbon, GHG, and water outcomes

Supplementary external observational data includes:

- Protected Areas Database of the United States (PAD-US)
- Cropland Data Layer (CDL) and Annual Crop Inventory for crop data in the U.S. and Canada
- Watershed boundaries from USDA Natural Resources Conservation Service
- SSURGO soil data from USDA Natural Resources Conservation Service
- Field boundaries from USDA Farm Service Agency

Describe (what, where, when) and list additional data (not captured in question above), plus samples, physical collections, software, curriculum materials and other materials produced during this period. Who is the expert or person(s) tasked with data collection (please provide their name and affiliation with the project)? This can be any member of the team. For data focused projects, that individual is expected to be key personnel.

There are no data produced in addition to that reported above.

Data collection has varied across ESMC team members (whose roles, titles, and tenure may have changed over the course of the project), and may include: Caroline Wade (Deputy Director), Austin Arrington (Technical Product Lead), Amanda Raster (Research Director), Benjamin Bartley (Standards & Protocol Director), Grant Ivison-Lane (Standards & Protocol Director), Jake Deutmeyer (Project Manager), Lizzie Rose (Project Manager), Stacy Cushenbery (Project Manager), and Neville Millar (Project Manager).

Provide how the data are organized and managed for all data created and utilized during this project. This could include the relevant data standardization, data formats, data definitions (e.g., data dictionary, ontologies), data organization, and/or where the data is stored. Please note if any of these data are confidential, and if so, when you would anticipate being able to share them publicly in the future.

Agronomic data collected from deployment and demonstration projects are entered into ESMC's digitized Monitoring, Reporting, and Verification (MRV) platform through either the web application's user interface (UI) or via ESMC's REST API. Application data is stored in a secure relational database, and producer identifying information is anonymized. Additional documents (i.e., shapefiles, soil data, producer contracts) are held in a secure cloud storage platform that is protected with two-factor authentication.

Quality Assurance/Quality Control (QA/QC) checks are performed on the data using automated software checks for hard and soft limits, as well as corroboration with satellite imagery data.

Agronomic data stored in the MRV is normalized to meet the needs of multiple technical partners. The data is normalized and exported to technical partners (via API integration) to model the GHG, carbon, and water footprint of the agronomic activities collected in the MRV at the field level.

Due to privacy policies and contractual agreements, ESMC does not share producer data with external parties. We will continuously share summaries of outcomes at an aggregated and anonymized level once project sites have been assessed and evaluated for program purposes.

FFAR supports the FAIR principles. We would like to understand how your project aligns with FAIR principles (for definition, see [Go Fair](#), for more information see [Nature Magazine](#)). Does your data align with the 4 principles in FAIR? Please check all boxes that apply and provide a brief description.

Not applicable

For each checked box, please provide a brief description.

Because ESMC collects producer data and personally identifiable information to quantify on-farm ecosystem services credits, we do not adhere to the FAIR principles in order to protect contractually-guaranteed producer privacy requirements.

What are the provisions for the data for the appropriate protection of privacy, confidentiality, security, intellectual property, or other rights or requirements?

All producer data, including personally identifiable information, is stored on a secure, web-based portal that is accessible only through multi-factor authentication login. Producer-specific credit generation outcomes are made available to each individual producer via a producer report, while buyers receive aggregated quantification results for each of their relevant projects. All third parties contracted on behalf of ESMC are required to sign a Data Use Agreement that prohibits them from using producer data

for purposes other than the project(s) for which their services are secured. Data in the form of reports and other program deliverables that is generated through ESMRC research projects (i.e., not producer or field/farm level data) is made available only to ESMC Members for one year after completion of the project/report, after which time it becomes publicly accessible and shared, to the extent applicable.

Describe the plans for archiving data, samples, and other research products, and for preservation of access to them. Refer to the FFAR Data Guidance Document for more information

All producer data, including personally identifiable information, is stored on a secure, web-based portal that is accessible only through multi-factor authentication login. Soil samples used to determine soil organic carbon stocks for quantification of credits are retained by the soil analysis laboratory providing services to each project, for a period of three to six months, after which time they are discarded. Research products developed by ESMC/ESMRC that are used in the credit quantification process, such as custom tools and quantification methodologies, are treated as the intellectual property of ESMC and are not made available to the public. All other research products developed by ESMC/ESMRC, such as reports, are made available only to ESMC Members, via Basecamp, for one year after completion of the project/report, after which time they become publicly accessible, if applicable, via ESMC's website, newsletter, and other forms of dissemination as previously noted.

Will this project coordinate with existing FFAR-supported grants and their data products? If so, provide a brief description of how, including the project title and PI of the existing FFAR work. If not, provide a brief rationale for why this project does not coordinate with existing FFAR supported work.

No. While we do coordinate with other FFAR-supported grants and projects (such as OpenTEAM), it is from a collaboration and information and research gap sharing approach rather than data or data product sharing. We collect and generate producer-level data, which is not appropriate for dissemination to external parties due to privacy concerns and contractual obligations to address those concerns. However, we freely and widely share other work accomplished and developed through the grant with other collaborators.

Please provide any feedback on these new data questions including any areas where you would like to see more clarity or explanation.

Not applicable

Certification

The undersigned hereby certifies that to the best of their knowledge and belief, the above report and all supporting documents are true, accurate and complete. I am aware there is a significant penalty for submitting false or misleading information.

If applicable, I agree that my electronic signature is legally binding, equivalent to, and has the same validity and meaning as my handwritten signature. I will not claim otherwise.

PI Full Name: _____

PI Signature: _____ Date: _____

Authorized Signing Official (ASO) Name: _____

ASO Signature: _____ Date: _____